

Fractal-Scaled Surface Velocity Thresholds and Idealized Fingerprint Curves for Pulsar Glitches in the Scalar-Angular-Torsion Framework

Nathan McKnight
Independent Researcher
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We present a deterministic, scale-invariant model of pulsar glitches within the Scalar-Angular-Torsion (SAT) framework. Pulsars are treated as macroscopic filamental bundles interacting with a discretized 4D lattice. Glitches arise when surface velocities exceed a universal critical fraction of the speed of light. Applying a fractal scaling factor from atomic thresholds to neutron-star systems, we define dimensionless surface velocity ratios and construct idealized fingerprint curves for spin jumps, glitch frequency, and luminosity modulation. This framework produces quantitative, falsifiable predictions for pulsar populations.

I. INTRODUCTION

Pulsar glitches are usually modeled stochastically via superfluid vortex dynamics or crustal shifts. Within the SAT framework, we propose a deterministic, geometric threshold mechanism. Surface layers temporarily disengage from electromagnetic interactions while retaining gravitational coupling, producing discrete jumps in spin and observable timing residuals.

Because the underlying forces operate fractally across scales, atomic-scale thresholds can be transferred to astrophysical systems using a scaling factor. Correctly applying this factor avoids prior overestimates of the velocity threshold.

II. VELOCITY-THRESHOLD MECHANISM

Let R denote the stellar radius and ω the angular velocity. The surface velocity is

$$v = \omega R. \quad (1)$$

Define the dimensionless ratio

$$x = \frac{v}{v_{\text{crit}}}, \quad (2)$$

where v_{crit} is the scaled critical velocity for 3D decoupling.

The fraction of mass participating in the decoupling is

$$f_{\text{decouple}}(x) = (1 - x^{-3}) H(x - 1), \quad (3)$$

with $H(x)$ the Heaviside function.

The corresponding spin jump is

$$\frac{\Delta\omega}{\omega} \approx f_{\text{decouple}}(x), \quad (4)$$

while the glitch rate is approximately

$$f_{\text{glitch}}(x) \sim \frac{d}{dx} f_{\text{decouple}}(x) = \frac{3}{x^4} H(x - 1). \quad (5)$$

The luminosity fraction of the outer layers is

$$L(x) \sim 1 - f_{\text{decouple}}(x) = \begin{cases} 1, & x < 1, \\ x^{-3}, & x \geq 1. \end{cases} \quad (6)$$

These relations produce the *pulsar fingerprint curves* for spin jump, glitch frequency, and luminosity modulation as functions of x .

III. IDEALIZED FINGERPRINT CURVES

Figure 1 shows the idealized curves for spin jump, glitch rate, and luminosity fraction. A sine-wave representation, scaled by the glitch-rate proxy, illustrates the discrete nature of individual glitch events.

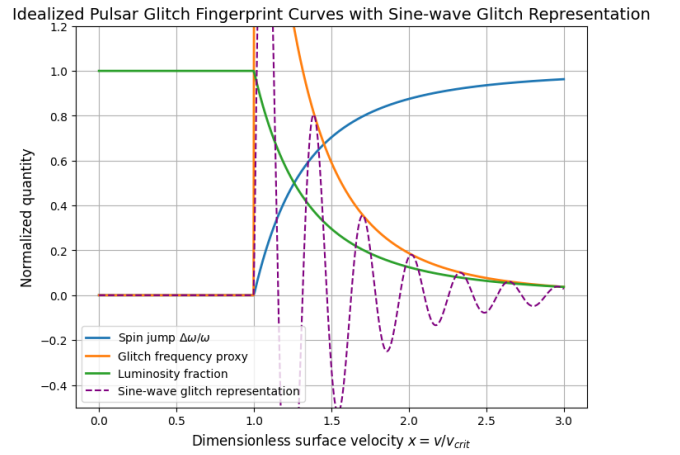


FIG. 1. Idealized SAT pulsar fingerprint curves: spin jump ($\Delta\omega/\omega$), glitch frequency proxy, luminosity fraction, and sine-wave glitch representation. The dimensionless surface velocity $x = v/v_{\text{crit}}$ governs the transition.

Key features:

1. Below threshold ($x < 1$): no glitches, normal luminosity.
2. Just above threshold ($x \gtrsim 1$): spin jumps onset, glitch frequency peaks.

3. High x ($x \gg 1$): spin jumps saturate, glitch frequency decreases as outer layers are fully decoupled.
4. Luminosity suppression coincides with threshold onset, stabilizing at a lower level as decoupling saturates.

IV. DISCUSSION AND PREDICTIONS

This model produces clear, falsifiable predictions:

1. Glitching pulsars cluster near $x \sim 1$.
2. Maximum spin jumps occur just above threshold; slower pulsars rarely glitch.
3. Glitch frequency and amplitude correlate with x , producing a population-level fingerprint.
4. Luminosity dips coincide with glitch onset, reflecting outer-layer decoupling from EM interactions.

The fractal scaling factor between atomic and astrophysical thresholds is critical to match observed surface

velocities. Once set, it constrains all predictions of spin jumps and glitch rates.

V. CONCLUSION

Integrating a fractal-scaled surface velocity threshold into the SAT framework yields a deterministic, falsifiable model for pulsar glitches. The fingerprint curves encode both spin-jump amplitude and glitch frequency in terms of a single dimensionless parameter. Application to the observed pulsar population will allow extraction of the empirical critical velocity and direct comparison with observations.

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